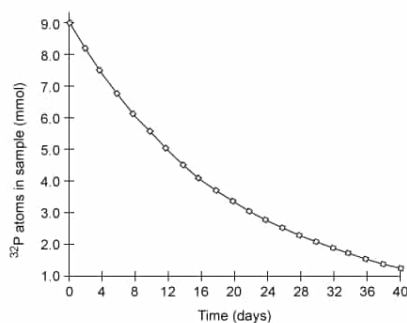


Phosphorus-32 is a radioactive beta-emitter used in the treatment of polycythemia vera (excess red blood cells). The radioactive decay profile for the number of ^{32}P atoms (millimoles) in a 2.0-mL sample was plotted, as shown below.



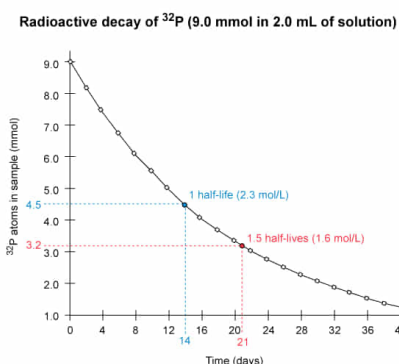
Based on the graph, what is the concentration (in mmol/mL) of ^{32}P in the 2.0-mL sample after 1.5 half-lives have passed?

- ☐ A. 1.2 mmol/mL [8%]
- ☒ B. 1.6 mmol/mL [27%]
- ☐ C. 2.3 mmol/mL [19%]
- ☐ D. 3.2 mmol/mL [43%]

Correct

28% answered correctly

Explanation:



In a **radioactive decay**, the **half-life** is the time required for an amount of a given isotope to **decrease by half**. With each subsequent half-life interval that passes, the amount remaining again decreases by half and the fraction of the initial amount that remains becomes **exponentially smaller**, eventually approaching zero. Accordingly, the fraction remaining after n half-lives can be calculated:

$$\text{Fraction remaining} = \frac{1}{2^n}$$

Following a second half-life interval ($n = 2$), the fraction remaining is again cut in half, making the resulting amount equal to one fourth the initial amount. Three half-lives ($n = 3$) would leave an amount equal to one eighth the initial amount.

In the graph of the radioactive decay of ^{32}P , the sample initially (0 days) has 9.0 mmol of ^{32}P atoms. The time required for this amount to decrease by 50% to 4.5 mmol is 14 days; therefore, the half-life of ^{32}P is 14 days. As such, a time interval of 1.5 half-lives is equal to 21 days.

$$1.5^\circ \text{half-lives} \times \frac{14 \text{ days}}{1^\circ \text{half-life}} = 21^\circ \text{days}$$

At 21 days (1.5 half-lives), the graph shows that there are 3.2 mmol of ^{32}P atoms in the sample. To determine the **molar concentration** (mol/L) of ^{32}P in the sample, the number of moles should be divided by the 2.0-mL sample volume:

$$\text{Molarity (M)} = \frac{\text{moles of analyte}}{\text{liter of solution}} = \frac{\text{millimoles of analyte}}{\text{milliliter of solution}} = \frac{3.2 \text{ mmol } ^{32}\text{P}}{2.0^\circ \text{ mL of solution}} = 1.6 \text{ mmol/mL}$$

Therefore, the concentration of ^{32}P in the sample after 1.5 half-lives is 1.6 M.

(Choice A) The solution has a ^{32}P concentration of 1.2 M after 2.0 half-lives have passed.

(Choice C) The solution has a ^{32}P concentration of 2.3 M after 1.0 half-life has passed.

(Choice D) The solution has a ^{32}P concentration of 3.2 M after 0.5 half-life has passed. A value of 3.2 mmol (with units of quantity, not concentration) is also obtained by failing to divide the number of millimoles by the sample volume in the last step of the calculation.

Educational objective:

A half-life is the time required for an amount of a given radioactive isotope to decrease by half. With each subsequent half-life interval that passes, the amount that remains decreases by half again, becoming exponentially smaller and eventually approaching zero. The molar concentration of a solution at a given time is determined by dividing the number of moles of analyte by the sample volume (in liters).

Time spent:0 sec

Subject:
General Chemistry

QID:402603

System:
Atomic Theory and Chemical Composition